

## Casimir Effect (quick)

- Idea is to renormalize  $\rho_0$  (vacuum energy density)

Dirac Eqn in mag field:

$$\rightarrow H = \frac{\vec{p}^2}{2m} + V(x) + \frac{e}{2m} \vec{B} \cdot (\vec{L} + g \vec{S})$$

$$w/ \vec{S} = 1/2 \vec{\sigma}$$

- $g$  is  $g$  factor electron
- strength of intrinsic dipole

- Dirac eqn

$$\rightarrow g = 2$$

- E2o premanent

$$g = 2 + \text{Lamb}$$

$\rightarrow$  anomalous mag moment

changed spinors

$$- (i \not{D} - m) \psi = 0$$

$$- (\not{p}^2 + m^2) \psi = 0$$

$$u \gg 0 \quad \not{D}^2 = \not{D}_m^2 + \frac{e}{2} \vec{F}_{m2} \cdot \vec{\sigma}$$

$$\rightarrow (\not{D}_m^2 + \frac{e}{2} \vec{F}_{m2} \cdot \vec{\sigma}) \psi = 0$$

diff between  
scalar & spinors

wegl:

$$e/m \text{ Dirac } \sigma^{42} = -e \left( (\vec{B} + i\vec{E}) \cdot \vec{\sigma} \right) / (\vec{B} - i\vec{E}) \cdot \vec{\sigma} \right)$$

paper

$$\rightarrow \frac{(H - e\vec{A})^2}{2m} \psi = \left( \frac{k^2}{2} + \frac{(\vec{P} - e\vec{A})^2}{2m} - 2 \frac{e}{2m} \vec{B} \cdot \vec{S} \pm i \frac{e}{2m} \vec{E} \cdot \vec{S} \right) \psi$$

- compare to other Hamiltonians

$\rightarrow g=2$  again

$$- \text{if } g' \in \text{Dirac } \sigma^{42} \rightarrow g=g'$$

correction

Consider a pf-shell system

$\rightarrow$  corrections add  
Dirac  $\sigma^{42}$  term

-  $\hat{p}, \hat{m}$  not coupled

process:

$$\bar{e}(q_2) \gamma_\mu (p) \rightarrow e^-(q_1)$$

$$\rightarrow \text{tree: } iM_i^* = \text{diagram} = -ie \bar{u}(q_2) \gamma^\mu u(q_1)$$

protonium is  $E_p M_0^*$

$\rightarrow g=2$  again...

coordon 1st order

$$\bar{u}(q_2) (q_1^\mu + q_2^\mu) u(q_1) = 2m \bar{u}(q_2) \gamma^\mu u(q_1) + i \bar{u}(q_2) \sigma^{\mu\nu} \times (q_1^\nu - q_2^\nu) u(q_1)$$

$$\rightarrow M_i^{\mu\nu} = -e \underbrace{\left( \frac{q_1^\mu + q_2^\mu}{2m} \right)}_{\text{non-relativistic } Q \rightarrow D} \bar{u}(q_2) u(q_1) - \underbrace{\frac{e}{2m} i \bar{u}(q_2) \gamma_2 \sigma^{\mu\nu} u(q_1)}_{\text{spin dep. \& giving mag moment}}$$

$$\rightarrow g = \frac{4\pi}{e} \times \text{coeff of } i \bar{p}_2 \bar{u} \sigma^{\mu\nu} u$$

-  $i \bar{u} \not{p}_2 \sigma^{\mu\nu} u$  modified using loops

General loop:

$$i M^{\mu} = \text{Diagram} = - \bar{u}(q_2) (f_1 \not{\gamma}^{\mu} + f_2 \not{p}^{\mu} + f_3 \not{q}_1^{\mu} + f_4 \not{q}_2^{\mu}) u(q_1)$$

- Every Feynman parameterized by  $f_i$

$$F_i = F(p, q, p^2, p, \dots)$$

momentum conservation  $\rightarrow p_2 = 0$

-  $F(q_1, q_2)$  can be reduced via

$$q_1^{\mu} u(q_1) = m u(q_1),$$

$$\bar{u}(q_2) q_2^{\mu} = m \bar{u}(q_2)$$

$$\rightarrow F_i \text{ only dep on } q_1, q_2, m^2 \text{ or } p^2 = 2m^2 - 2q_1 \cdot q_2 + f_{22}$$

using identity:

$$0 = p_m \bar{u} (f_1 \not{\gamma}^{\mu} + f_2 \not{p}^{\mu} + f_3 \not{q}_1^{\mu} + f_4 \not{q}_2^{\mu}) u$$

$$= f_1 \bar{u} \not{p} u + (p \cdot q_1) f_3 \bar{u} u + (p \cdot q_2) f_4 \bar{u} u$$

$$= (p \cdot q_1) f_3 \bar{u} u + (p \cdot q_2) f_4 \bar{u} u$$

$$p \cdot q_1 = q_1 \cdot q_1 - m^2 = -p \cdot q_2$$

$$f_3 = f_4$$

$$\therefore i M^{\mu} = (-ic) \bar{u}(q_2) \left( F_1 \left( \frac{p^2}{m^2} \right) \not{\gamma}^{\mu} + \frac{i \sigma^{\mu\nu}}{2m} p_2 F_2 \left( \frac{p^2}{m^2} \right) \right) u(q_1)$$

-  $F_1, F_2$  from Pade's

- breaking gives  $F_1 = 1, F_2 = 0$

-  $F_1$  modifies  $\mathcal{L}_{\text{int}} \bar{\psi} \psi \phi$  coupling

-  $F_2$  gives my moment

-  $F_2 = 0 \rightarrow g = 2$   
 $F_2 = F_2(\frac{p^2}{m^2}) \therefore g \rightarrow 2 + 2F_2(\frac{p^2}{m^2})$

- Wilson's thing,  $\epsilon \rightarrow p \ll m$ :

$$g = 2 + 2F_2(0)$$

graphical:

1)



2)



3)



- only counter. b/w to  $F_1$

4)  $iM^m =$

-  $pk = qk - qk$

- For  $g = 2$ , only graph needed

comp. by Feynman rules:

$$iM^m = (-ie)^3 \int d^4k \frac{-ig^2 \alpha}{(k-q)^2 + i\epsilon} \bar{u}(q_2) \gamma^2$$

$$\times \frac{i(\not{p} + \not{q} + m)}{(p+k)^2 - m^2 + i\epsilon} \gamma^m \frac{i(\not{k} + m)}{k^2 - m^2 + i\epsilon} \not{d} u(q_1)$$

$$= -e^3 \bar{u}(q_2) \int d^4k \frac{\gamma^2 (\not{p} + \not{k} + m) \gamma^m (\not{k} + m) \not{d}}{[(k-q)^2 + i\epsilon] [(p+k)^2 - m^2 + i\epsilon] [k^2 - m^2 + i\epsilon]} u(q_1)$$

using

$$\frac{1}{ABC} = 2 \int_0^1 dx dy dz \delta(x+y+z-1) \frac{1}{[xA+yB+zC]^3}$$

$$w/ \quad A = k^2 - m^2 x_1^2$$

$$B = (p+m)^2 - m^2 x_1^2$$

$$C = (a-x_1)^2 + x_1^2$$

$$\rightarrow xA + yB + zC = k^2 + 2xy p \cdot (x_1) + y p^2 + z x_1^2 - (xy) k^2 x_1^2 \\ = (k^2 + y p^2 - 2x_1^2) - \Delta x_1^2$$

$$w/ \Delta = -2y p^2 + (1-z) m^2$$

$$\text{shift } k^\mu \rightarrow k^\mu - y p^\mu + z q^\mu$$

$$\text{Denom now } (k^2 - \Delta)^3$$

Numerator:

$$N^\mu = \bar{u}(q_2) \gamma^\mu (\not{p} + \not{p} + \not{m}) \gamma^\mu (\not{p} + \not{m}) \gamma^\mu u(q_1)$$

$$= 2 \bar{u}(q_2) [\not{p} \not{p} \not{p} + \not{p} \not{p} \not{p} + \not{p} \not{p} \not{p} - m(2m + p^\mu)] u(q_1)$$

$$k^\mu \rightarrow k^\mu - y p^\mu + z q^\mu$$

implies

$$-\frac{1}{2} N^\mu = \bar{u}(q_2) [(p - y p + z q) \gamma^\mu + (p - y p + z q) \gamma^\mu (p - y p + z q)] u(q_1)$$

$$+ \bar{u}(q_2) [m^2 \gamma^\mu - m(2m - 2y p^\mu + 2z q_1^\mu + p^\mu)] u(q_1)$$

$$\text{using } m^2 = \frac{1}{4} g^{\mu\nu} m^2, \text{ g.o.s. ident, } x+y+z=1,$$

obtain

$$-\frac{1}{2} N^\mu = \left[ -\frac{1}{2} k^2 + (1-x)(1-y) p^2 + (1-4z+z^2) m^2 \right] \bar{u}(q_2) \gamma^\mu u(q_1)$$

$$+ i m z (1-z) p_2 \bar{u}(q_2) \gamma^{\mu} u(q_1)$$

$$+ m (2-z) (1-z) p^{\mu} \bar{u}(q_2) u(q_1)$$

-  $p^{\mu}$  / would not contribute to loop contr.

$$- i M_{\pi}^{\mu} (p^{\mu} \text{ term}) = 4e^2 \int_0^1 dx dy dz \delta(x+y+z-1) m(z-z) \bar{u}(q_2) u(q_1) \left( \int_0^1 dx \frac{p^{\mu}}{(k^2 - \Delta + i\epsilon)^3} \right)$$

- Integral common in many

- Integral measure symm. in many  $\rightarrow$  consider  $\uparrow$

$$\therefore i M_{\pi}^{\mu} = p_2 \bar{u}(q_2) \gamma^{\mu} u(q_1) \left[ 4e^2 m \int_0^1 dx dy dz \delta(x+y+z-1) \times \int_0^1 dx \frac{2(1-z)}{(k^2 - \Delta + i\epsilon)^3} \right] \text{ rest contributes to } g$$

-  $F_2(p^2)$  is cft  $\times \frac{2m}{e}$

$$\Rightarrow F_2(p^2) = \frac{2m}{e} (4e^2 m) \int_0^1 dx dy dz \delta(x+y+z-1)$$

$$\times \int_0^1 dx \frac{2(1-z)}{(k^2 - \Delta + i\epsilon)^3} + O(e^4)$$

(additionally,  $F_1(p^2) = 1 + O(p^2) + O(e^4)$ )

$$w/ F(p^2) = -2ie^2 \int_0^1 dx dy dz \delta(x+y+z-1)$$

$$\times \frac{k^2 - 2(1-z)(1-y)p^2 - 2(1-4z+z^2)m^2}{[k^2 - (m^2(1-z)^2 - m_j^2 p^2)]^3}$$

$$F_2: \int_0^1 dx \frac{1}{(k^2 - \Delta + i\epsilon)^3} = \frac{-i}{32\pi^3} \left( \int_0^1 dx \frac{1}{[k^2 - (m^2(1-z)^2 - m_j^2 p^2)]^3} \right)$$

Then,  $F_2(p^2) = \frac{\alpha}{\pi} m^2 \int_0^1 \int_0^1 dx dy dz \delta(x+y+z-1) \frac{z(1-z)}{(1-z)^2 m^2 - xy p^2}$

$p^2=0$ ,

$$\begin{aligned} F_2(0) &= \frac{\alpha}{\pi} \int_0^1 dz \int_0^1 dy \int_0^1 dx \delta(x+y+z-1) \frac{z}{(1-z)} \\ &= \frac{\alpha}{\pi} \int_0^1 dz \int_0^{1-z} dy \frac{z}{(1-z)} \\ &= \frac{\alpha}{2\pi} \end{aligned}$$

$\therefore \left[ g = 2 + \frac{\alpha}{\pi} + \mathcal{O}(\alpha^4) \right]$